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Instructions for Installing Fuel Shutoff Solenoid Overlay Harness

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Read all instructions before beginning installation.

There are two harness part numbers available.

Table 1: Fuel Shutoff Solenoid Overlay Harness Options

Part Number	Description
3935750	For use with 30 amp fuse systems.
3935751	For use with 60 amp fuse systems.

Tools and Materials Needed for Installation:

- Wire cutters and strippers
- Heat gun (for shrink tubing)
- Terminal hand crimper - Recommended AMP 69355, or equivalent
- Electric drill and 6-mm [1/4-in] drill bit
- 8-mm socket and box/open-end wrench
- Torque wrench
- Harness mounting clips, wire ties, or appropriate harness attaching hardware
- Electrical tape
- Voltmeter or multimeter.

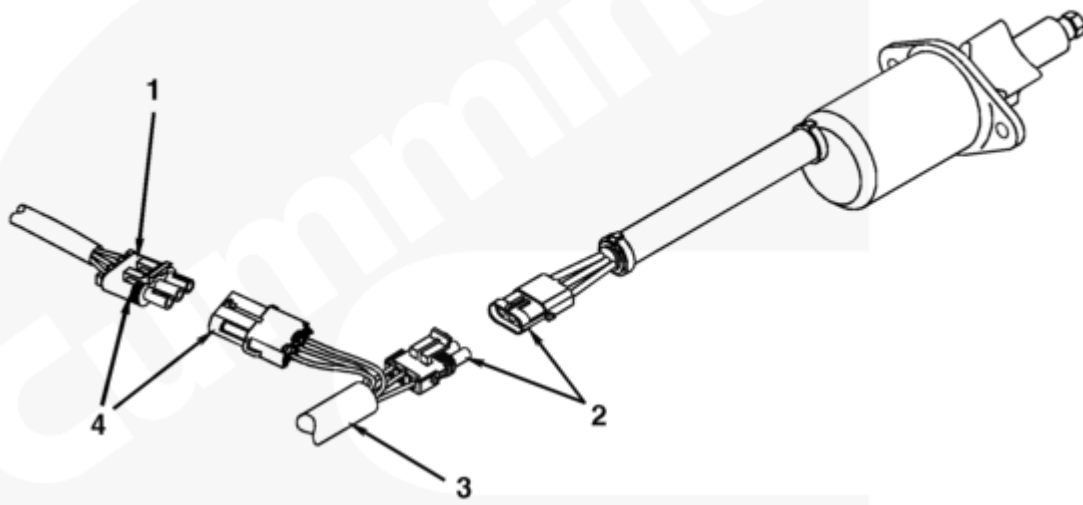
Table 2: Recommended Relay Mounting Hardware

Part Number	Description	Quantity
3901865	M6 x 1 x 20 hex flange head capscrews	2
3906216	M6 x 1 hex flange nuts	2

This harness is designed to provide sufficient voltage to operate the fuel pump shutoff solenoid. The harness is installed between the fuel pump solenoid and the existing OEM connection to that solenoid. The solenoid will receive its power from this harness and directly from the vehicle

battery. This harness is designed to allow the installer to size the battery-to-relay section to fit the vehicle. The relay-to-fuel pump sections are **not** adjustable. If the fuel shutoff solenoid is controlled by a timer, contact a Cummins® Authorized Repair Location.

1. Measure the battery voltage at the battery during the starting cycle. There **must** be a minimum of 11-VDC available. If the voltage is **not** at least 11-VDC, check the battery and charging system and charge or repair, as necessary. Also, check the fuse capacity on the existing fuel shutoff solenoid switching circuit. If the fuse is rated over 5 amps, it can possibly need to be replaced. The relay in the overlay harness draws very little amperage.
2. Determine the location for mounting the fuel shutoff solenoid relay. Make sure that the Packard™ harness connectors will reach the original fuel shutoff solenoid Packard™ connection. The relay **must** be mounted to a rigid surface under the hood and out of the way of road spray. Do **not** mount the relay on the engine. Tighten the mounting capscrews to the appropriate torque given for the surface the relay is mounted on.
3. Determine the best routing path from the relay to the battery. Leave enough length in wiring to avoid sharp bends. Route around any high temperature areas and moving parts. Secure with harness mounting clips, wire ties, or appropriate harness attaching hardware when routed correctly.
4. After routing the harness, cut the 8-AWG red and black leads to length at the battery. Remove excess convoluted tubing, if necessary. Strip approximately 9.5-mm [3/8-in] of the insulation away from the wire, and slide one piece of heat shrink tubing over each wire. Crimp the provided 9.5-mm [3/8-in] ring terminals onto each wire. Apply heat to the heat shrink tubing on the crimped portion of the terminal. Make sure that all wire strands are covered. Reattach the convoluted tubing to the 8-AWG leads with electrical tape.
5. Disconnect the existing electrical connection from the fuel shutoff solenoid (2). Connect the fuel shutoff solenoid to the 3-way Weather Pack™ tower connector on the overlay harness (3). Connect the OEM 3-way Weather Pack™ tower connector to the 3-way Weather Pack™ shroud connector on the overlay harness (1). See Figure 1.



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Fig. 1 Overlay Harness Connection

1. Existing OEM connector
2. New solenoid connection
3. Overlay harness
4. New OEM connection.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

After all engine compartment connections have been made and the relay is properly mounted, connect the 8-AWG red and black leads to the battery.

The installation is complete.

Document History

Date	Details
xxx	Module Created
2012-11-7	Two part numbers, 3935750 and 3935751, must be added corresponding to the 30 amp and 60 amp fuel model overlay harnesses, respectfully.

